

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Dinnepati Sindhu Prasad
Register Number	192311271

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Dinnepati Sindhu Prasad-192311271 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dinnepati Sindhu Prasad

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Q1. Cloud Architecture for a Video Streaming Platform

Answer:

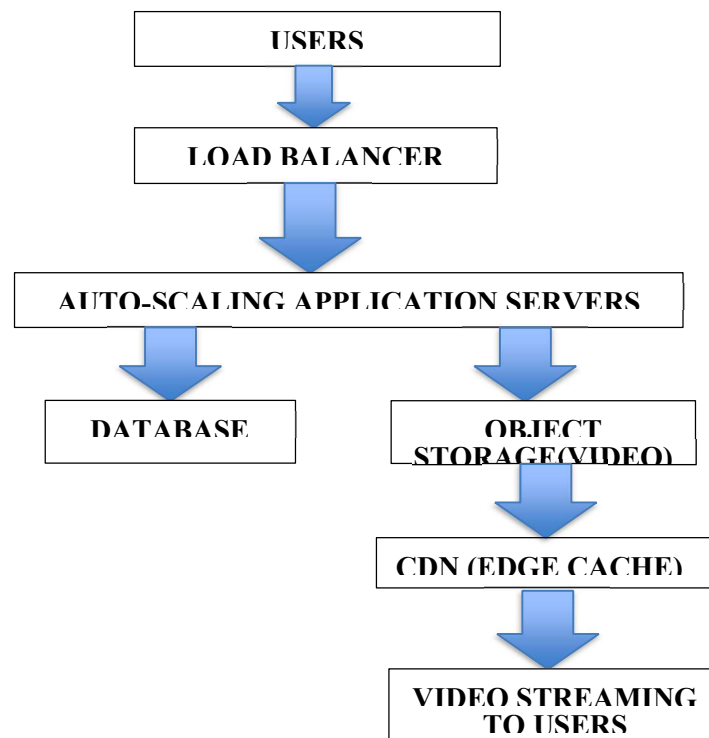
1. Cloud Architecture Design

To support millions of users watching videos simultaneously, the media company should use a scalable cloud architecture that automatically adjusts resources based on demand. A **Content Delivery Network (CDN)** stores frequently accessed videos at edge locations to reduce buffering, while **Load Balancers** and **Auto Scaling Groups** ensure that traffic is distributed efficiently among multiple application servers.

The architecture consists of the following components:

- Users access the streaming service using web or mobile applications.
- A Global Load Balancer distributes incoming requests across available servers.
- Auto Scaling Groups automatically launch or terminate application servers depending on traffic.
- A CDN caches popular videos closer to users for faster delivery.
- Cloud Object Storage stores original video files.

Architecture Flow:



3. Role of Load Balancing and Auto-Scaling

Load Balancing

A Load Balancer plays an important role in handling a large number of user requests.

- It distributes requests evenly among all application servers.
- It prevents any single server from becoming overloaded.
- If one server fails, traffic is automatically redirected to healthy servers.
- It improves application availability and provides a better user experience.

Auto-Scaling

Auto Scaling adjusts computing resources automatically according to user demand.

- During peak hours, new application servers are created automatically.
- During low traffic periods, unnecessary servers are removed.
- It maintains consistent streaming quality.
- It minimizes buffering even when millions of users access videos simultaneously.

4. Cost Optimization Strategies

The company can reduce operational costs using the following methods:

- Store video files in cost-effective cloud object storage.
- Cache frequently viewed content using a CDN to reduce bandwidth usage.
- Enable Auto Scaling so that resources run only when required.
- Use Reserved Instances for predictable workloads and Spot Instances for video encoding jobs.
- Compress videos using modern codecs such as H.265 or AV1 to reduce storage and network costs.
- Apply lifecycle policies to move older videos to archival storage.
- Continuously monitor resource usage and remove idle resources.

Conclusion

A cloud-based video streaming platform designed with Load Balancing, Auto Scaling, CDN caching, and Cloud Storage can efficiently support millions of concurrent users while maintaining smooth video playback. These technologies improve scalability, reduce buffering, ensure high availability, and optimize cloud infrastructure costs.

2.A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Answer:

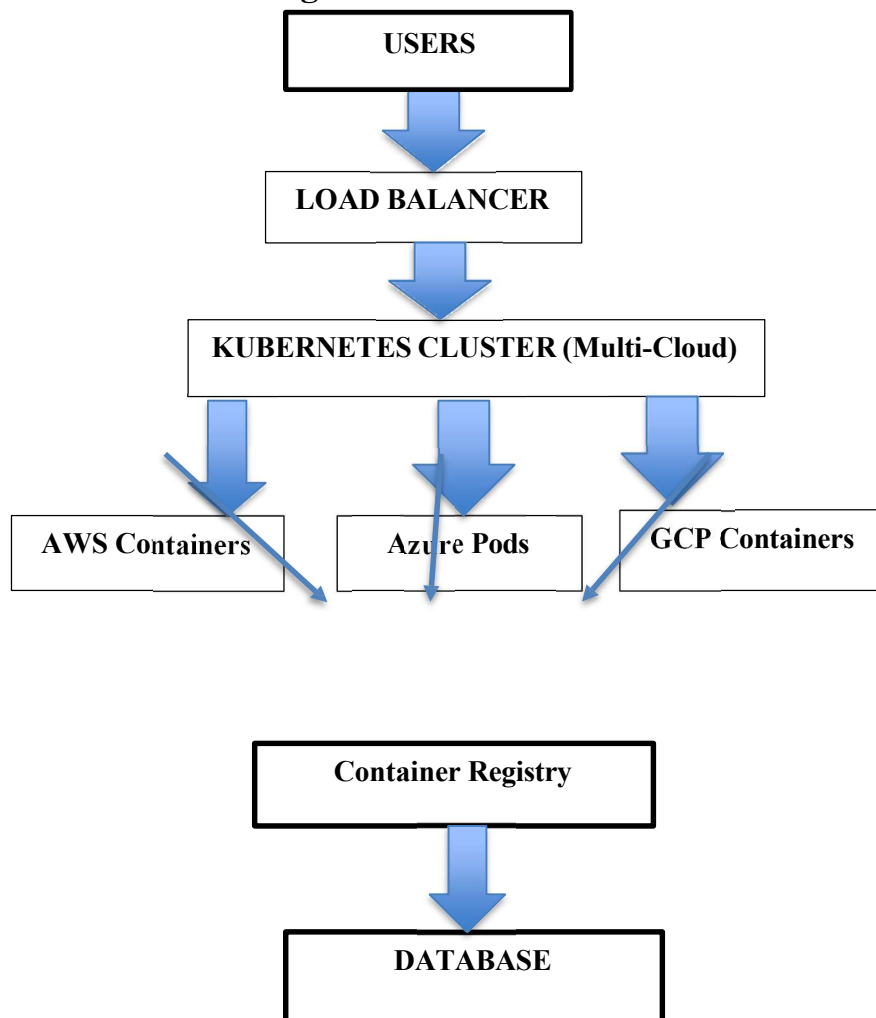
1. Proposed Solution

To maintain consistent application deployments across different cloud providers, the fintech startup should use Docker for containerization and Kubernetes for container orchestration. Kubernetes allows applications to run uniformly on AWS, Microsoft Azure, and Google Cloud without changing the application code.

The proposed solution includes:

- Package each application as a Docker container.
- Deploy containers on Kubernetes clusters in multiple cloud providers.
- Store container images in a centralized Container Registry.
- Use a CI/CD pipeline to automate testing and deployment.
- Use a cloud load balancer to distribute user traffic

1. Architecture Diagram



3. Service Discovery and Scaling

Service Discovery

Kubernetes automatically manages communication between containers.

- Each service receives a unique DNS name.
- Applications communicate using service names instead of IP addresses.
- Failed containers are automatically replaced.

Scaling

Kubernetes provides automatic scaling using:

- Horizontal Pod Autoscaler (HPA)
- Cluster Autoscaler
- Load balancing across available pods

This ensures the application continues to perform efficiently during varying workloads.

4. Benefits and Risks of Multi-Cloud

Benefits

- High availability
- Better disaster recovery
- Avoids vendor lock-in
- Improved global performance
- Flexibility to use different cloud services

Risks

- Complex infrastructure management
- Higher networking costs
- Data synchronization challenges
- Security policy management across providers
- Difficult monitoring and troubleshooting

Conclusion

Using Docker and Kubernetes across multiple cloud providers helps maintain deployment consistency, improves scalability, and increases application availability. Although managing multiple clouds requires additional effort, it provides greater flexibility and reliability.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

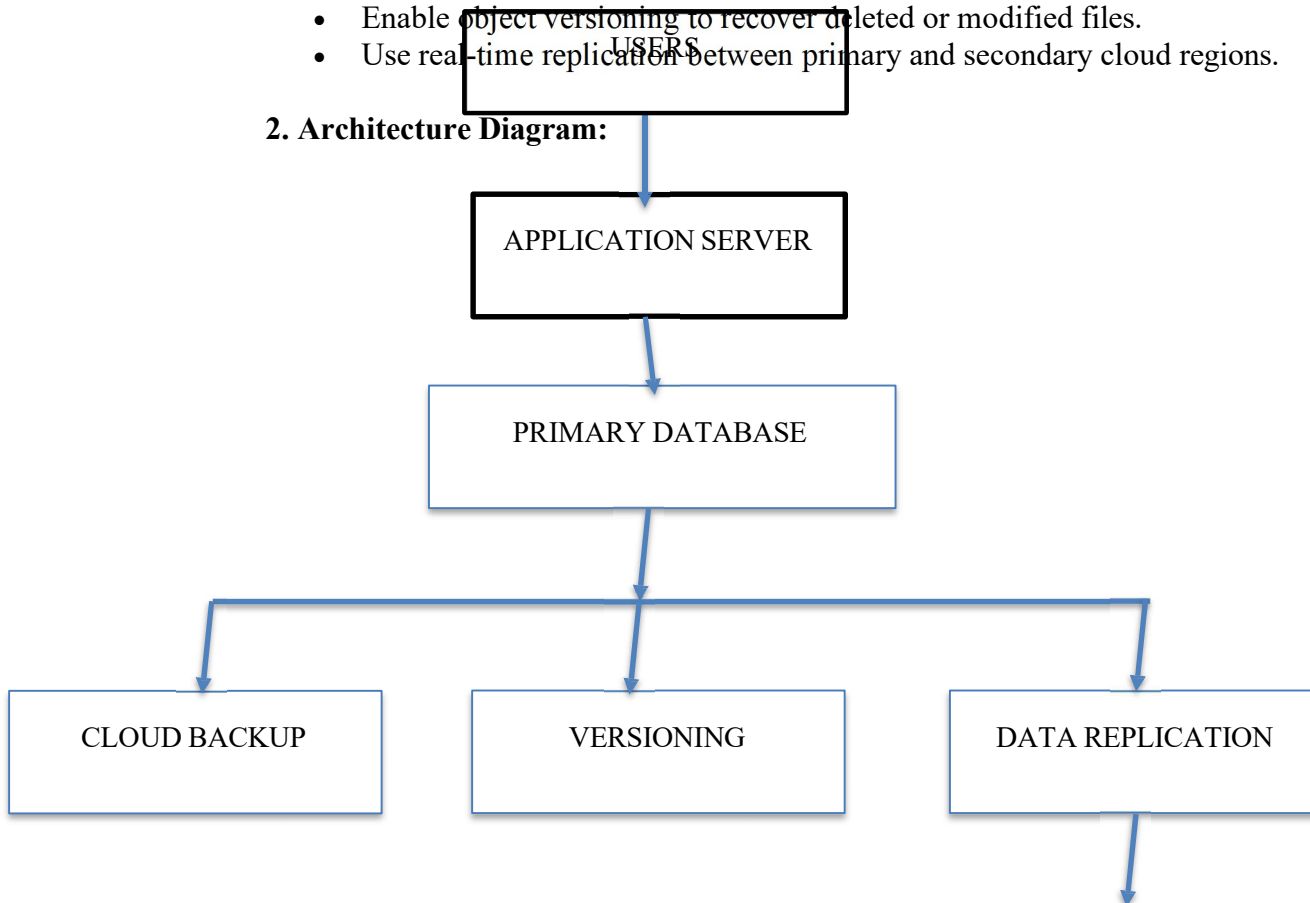
Answer:

1. Solution Design

The e-learning platform should use **cloud-native backup, data replication, and object versioning** to protect data and recover quickly from accidental deletion or system failures.

- Enable automatic cloud backups for databases and files.
- Store backup copies in multiple cloud regions.
- Enable object versioning to recover deleted or modified files.
- Use real-time replication between primary and secondary cloud regions.

2. Architecture Diagram:



SECONDARY REGION
(DR)



DISASTER RECOVERY

3. Achieving

RTO and RPO

Recovery Time

Objective

(RTO):

- Use automatic failover to the secondary region.
- Keep backup servers ready for quick recovery.
- Minimize downtime by restoring services within minutes.

Recovery Point Objective (RPO):

- Enable continuous data replication.
- Perform frequent automatic backups.
- Use versioning to restore the latest valid data with minimal data loss.

4. Steps to Ensure Business Continuity:

- Schedule automatic daily and incremental backups.
- Enable object versioning for all important files.
- Replicate data to another cloud region.
- Regularly test backup restoration and disaster recovery plans.
- Configure automatic failover for critical services.
- Continuously monitor backup status and system health.
- Maintain documented disaster recovery procedures.
- Encrypt backup data and apply access controls for security.

Conclusion:

By using cloud-native backups, data replication, object versioning, and disaster recovery services, **the e-learning platform can recover quickly from accidental deletion, achieve low RTO and RPO, and ensure uninterrupted learning services during failures.**

2.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Answer

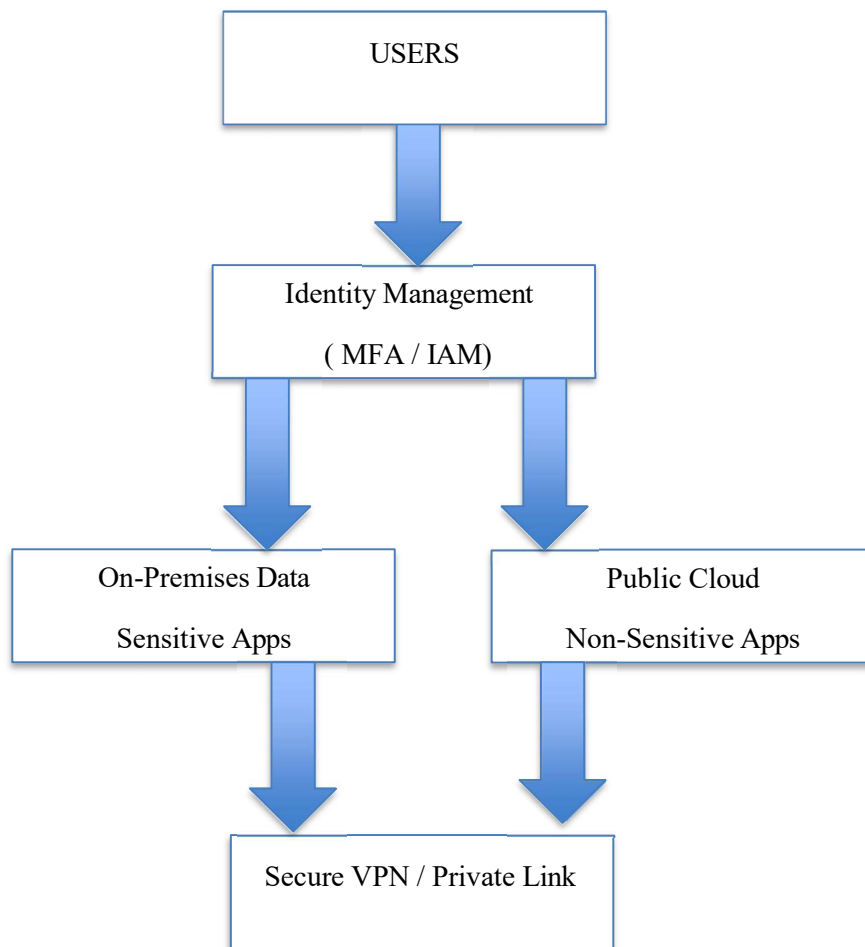
1.Hybrid Cloud Solution

A hybrid cloud allows sensitive applications to remain on-premises while non-sensitive workloads run in the public cloud.

The architecture includes:

- On-premises data center
- Public cloud services
- Secure VPN or dedicated private connection
- Identity and Access Management (IAM)
- Encryption for all data

2.Architecture Diagram



3.Identity Management and Encryption

- Identity and Access Management (IAM)
- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Encryption

- AES-256 encryption for stored data
- TLS/SSL encryption during transmission
- Secure Key Management Service (KMS)

Challenges in Hybrid Cloud

- Managing two different infrastructures
- Maintaining security consistency
- Compliance requirements
- Data synchronization
- Increased networking cost
- Monitoring multiple environments

Conclusion

A hybrid cloud provides both security and scalability by keeping confidential workloads on-premises while using cloud resources for less sensitive applications.

5)A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud

Answer

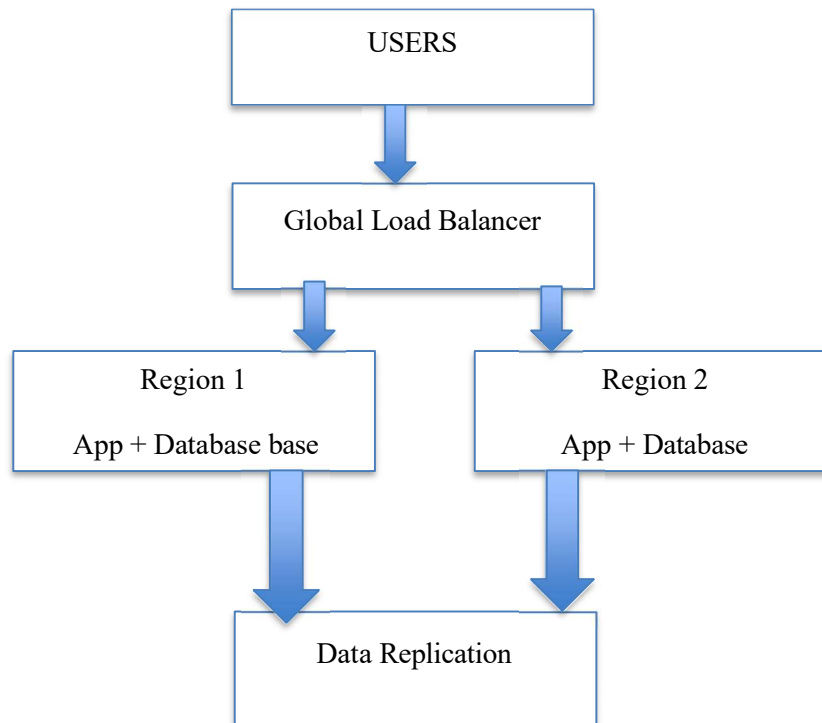
1.Multi-Region Deployment Strategy

To achieve high availability, the SaaS application should be deployed in multiple cloud regions.

The deployment includes:

- Global Load Balancer
- Multiple application servers
- Regional databases
- Continuous data replication
- Automatic failover

2.Architecture Diagram:



3.Data Replication

and Failover Data

Replication

- Real-time database replication
- Automatic snapshots
- Regular backups
- Synchronized application data

Failover

- Health checks monitor each region.
- Traffic automatically moves to a healthy region.
- Services continue without interruption.
- Failed regions are restored after recovery.

4.Monitoring and Alerting

Effective monitoring should include:

- CPU and memory utilization
- Network traffic
- Application response time
- Server health checks
- Centralized log collection
- Email/SMS alerts
- Regular disaster recovery testing

Conclusion

Deploying applications across multiple cloud regions with continuous replication, automatic failover, and proactive monitoring ensures high availability and fault tolerance. Even if one cloud region becomes unavailable, users continue to access the application without significant service interruption.